

Redwan Khalifa

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EDUCATION

Expected – 2026 Bachelor of Engineering (BEng) Computer Engineering Toronto Metropolitan University - Toronto, ON

- Minor in Computer Science, Mathematics, Biology and certification in ASL (American Sign Language).
- Relevant Coursework: RF Circuits, Digital Electronics, Embedded Systems, Microprocessors, Computer Architecture.

EXPERIENCE

06/2025 – Present Hardware/Embedded Systems Engineer MetRocketry - Toronto, ON

- Designed and assembled multi-layer modular PCBs in KiCad/Altium for telemetry, power delivery, and communication systems, integrating SMD components, sensors, and microcontroller bus protocols.
- Developed embedded C/C++ firmware to enhance communication reliability, performance, and modularity across flight subsystems.
- Implemented a GUI with database integration and CI/CD automation for real-time data visualization, validation, and telemetry monitoring.
- Debugged PCB-based avionics systems, optimizing signal integrity and modular communication between GPS, radios, and microcontrollers.

09/2025 – Present Algorithms and Systems Engineer TMU - Smart Campus Navigation System - Toronto, ON

- Designed and implemented backend architecture using Node.js, Express.js, and MongoDB for real-time indoor navigation and building data.
- Applied Dijkstra's algorithm for shortest-path routing, enabling multi-floor navigation and optimized query performance across connected maps.
- Integrated Firebase Authentication for secure user access and database protection.

03/2025 – 05/2025 Software Engineer Riipen - Toronto, ON

- Developed a responsive website for a trucking company in HTML, CSS, and jQuery to improve user experience, and engagement.
- Implemented a contact form with backend email integration for better communication with clients.
- Optimized metadata and site structure for mobile use to boost search visibility and user accessibility.

PROJECTS

Ongoing Project FPGA Memory Controller Design and Verification [GitHub Project Link](#)

- Designing a simplified cache controller for a CPU memory hierarchy using VHDL in Xilinx ISE.
- Currently implementing cache organization with tag, index, offset, valid, and dirty bits for block replacement logic.
- Targeted for deployment on Xilinx Spartan-3E FPGA, with plans for functional simulation and hardware emulation.

08/2025 Digital Signal Processing Filter Design and Audio Equalizer [GitHub Project Link](#)

- Implemented DTFT, DFT, and FFT algorithms in Python to analyze discrete signals and evaluate spectral properties.
- Designed, simulated, and tested an FIR high-pass filter; verified frequency response through spectral analysis.
- Built a multi-band audio equalizer (bass, mid, treble) demonstrating applied DSP techniques in real-world audio systems.

12/2023 Autonomous Maze Navigating Robot [GitHub Project Link](#)

- Programmed an eebot robot in HCS12 Assembly to autonomously navigate a maze using sensor input, state machines, and intersection logic.
- Developed custom subroutines for LCD display, analog-to-digital conversion (ADC), motor control, turn timers, and LED guidance system.
- Tuned sensor thresholds and logic flow to handle S-turns, dead ends, and bumper-triggered events through trial-and-error learning.

SKILLS

Hardware: FPGA/SoC (Quartus, Xilinx ISE), PCB Design (KiCad, Altium), Arduino, Raspberry Pi, HCS12, AutoCAD.

Programming: Python, C/C++, Verilog, VHDL, Java, MATLAB, Bash.

Lab: Oscilloscopes, Logic Analyzers, Spectrum Analyzers, Multimeters, PCB prototyping, soldering Iron.

OS & Tools: Windows, Linux, MacOS, Android, iOS, Git, Jupyter Notebook.